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Quantitative analysis of angular momentum generation in the triple loop in figure skating

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ABSTRACT

Whole-body Angular momentum (AM) is a key factor for jump execution and landing stability in figure skating. However, its generation mechanism in the triple loop (T-LP) remain underexplored. The aim of this study was to address this gap by analysing the segmental contributions to angular momentum generation. Ten female skaters performed the T-LP while undergoing three-dimensional motion analysis. Although the AM increased first due to the contribution of the upper limbs during the glide phase, the AM of the lower limbs increased more significantly during the transition phase, with a particularly notable contribution from the non-supporting leg. A slight reduction in AM was observed during the Pivot phase just before take-off. These findings suggest that, unlike off-ice jumps where AM is typically generated by the arms and trunk, T-LP on ice involves a different strategy. Furthermore, no significant correlation was found between AM at take-off and key jump performance parameter such as jump height, rotational velocity, or flight time. This suggests that optimal jump execution requires coordinated integration of other biomechanical factors, including take-off mechanics and aerial posture control. These insights may aid in refining training strategies and optimising jump mechanics in figure skating.

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Edge jump; motion capture; segmental contribution; jump mechanics; kinematics analysis

Introduction

Figure skating features six primary jump types: toe-loop, salchow, loop (LP), flip, lutz, and axel. The difficulty and point value of each jump are determined by the number of rotations (International Skating Union, 2024a). In recent years, both male and female skaters have actively incorporated triple and quadruple jumps into their routines to maximise their scores. In response to this technical progression, the International Skating Union (ISU) introduced a scoring scale for quintuple jumps for the first time in the 2024–2025 season (International Skating Union, 2024b). As the technical demand of this sports continue to evolve, understanding the biomechanical characteristics of each jump types is essential for both technical development and performance optimisation.

The axel is widely regarded as the most technically demanding jump due to its forward take-off, which requires an additional half turn when compared to other

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jumps (Vescovi & VanHeest, 2018). Numerous studies have examined the mechanics of the axel and have consistently found that increasing rotational velocity during flight is crucial for achieving multiple rotations (Albert & Miller, 1996; King et al., 1994). Rotational velocity during flight is primarily influenced by two key factors: angular momentum and moment of inertia. While skaters can reduce the moment of inertia by pulling their arms and legs inward, this strategy has physical limitations. However, rotational velocity can still be increased by generating greater angular momentum at take-off, even when the moment of inertia remains constant. Therefore, clarifying the mechanisms and kinematic factors contributing to angular momentum generation during the take-off sequences could provide valuable insights for optimising the jump technique. Albert and Miller (1996) examined angular momentum generation during the take-off sequences of single and double axels performed by 16 skaters. They divided the take-off sequences into three phases: the Glide phase, which begins when the supporting leg contacts the ice and ends when the centre of mass (COM) reaches its lowest point; the Transition phase, which starts as the COM begins to rise and continues until the skate rocks forward towards the toe-pick and makes contact with the ice; and the Pivot phase, which begins when the skater shifts their weight onto the toe-pick and ends at take-off. The findings indicated that the forward swing of the free leg during the later Glide and early Transition phases significantly contributed to angular momentum generation. However, the mechanics underlying other kinds of jumps are still unknown because their study focused solely on double axel jumps.

The LP is often referred to as 'The King of Jumps' due to its fundamental role in developing stability and balance in aerial posture across various jump types (Shulman, 2001; Wilkes, 1999). Since its introduction by Werner Rittberger in the early 20th century, the LP has continuously evolved, with Dick Button successfully landing the first triple loop (T-LP) at the 1952 Olympics. Over the decades, the T-LP has become a staple among elite-level skaters and is frequently incorporated into competitive programmes. Despite its technical significance and widespread use, biomechanical research on the LP remains limited. Existing studies have primarily focused on flight time (Ikegami et al., 2004) and jump height (Sakurai et al., 1999), but the mechanisms underlying angular momentum generation in the LP remain unclear. Given the critical role of angular momentum in rotational performance, further investigation is required to clarify how different body segments contribute to its generation during T-LPs. Therefore, this study aimed to (1) investigate the kinematic characteristics of the T-LP by comparing its fundamental kinematic variables with those of other jumps reported in previous studies and (2) Quantify the contributions of individual body segments to angular momentum generation during the take-off sequences, thereby providing insights into optimising take-off mechanics. Based on these aims, we hypothesised that the lower limbs would play the primary role in angular momentum generation during the take-off sequences of the T-LP, and that the segmental contributions would differ depending on the phase (Glide, Transition, and Pivot).

Materials and methods

Participants

This study included 10 collegiate and high school female figure skaters (age: 20.5 ± 2.7 years, height: 1.564 ± 0.032 m, mass: 53.2 ± 3.5 kg, skating experience: 14.6 ± 2.9 years). All participants had attained Grade 7 on the Japan Skating Federation Figure Skating Tests, a level requiring proficiency in at least two types of triple jumps and an advanced understanding of jump techniques. They all had competed at the national level, and nine of the ten skaters had also participated in international competitions at the junior or senior level. Furthermore, all participants had successfully landed a T-LP in competition during the current season. All skaters took off with their right foot and rotated counterclockwise. Prior to the experiment, participants were informed of the purpose of the study and experimental procedures, and written informed consent was obtained. This study was approved by the institutional ethics committee (protocol code: 2022–24).

Design and procedures

Participants performed the T-LP with maximum effort following a warm-up, continuing until they successfully executed a fully rotated jump. A successful attempt was defined as a fully rotated T-LP with no under-rotation or landing errors, as determined through video analysis and assessed by an expert with over 10 years of figure skating experience. The data collection area was set on half of the ice rink, covering a 15 m × 15 m capture space. This size was determined on the basis of the specifications of the motion capture system to ensure the highest accuracy within this range. The global coordinate system was defined with the X-axis as the lateral direction (rightward relative to the skater's initial motion), the Y-axis as the direction of travel (forward), and the Z-axis as the vertical direction (upward) (Figure 1A). The positions of 39 reflective markers placed on anatomical landmarks (Figure 1B) were recorded during experimental trials using a VICON, MX-T-20S motion capture system (Vicon Motion System, UK) equipped with 19 cameras operating at 200 Hz. To identify take-off and landing, a high-speed video camera (Fastec TS-3, Fastec Imaging, Figure 1A,B) captured the skater's feet at 200 Hz. The motion capture and high-speed video systems were electronically synchronised. Additionally, a video camera (Handycam, Sony, Figure 1A-C) recorded the trials at 60 Hz. Three-dimensional coordinate data were filtered using a Butterworth low-pass digital filter with a cut-off frequency of 15 Hz to reduce noise and enhance signal clarity. This cut-off frequency was chosen to minimise noise without excessive smoothing and is consistent with previous studies using data sampled at similar frequencies (King et al., 2018).

Analysis phases

The analysis phases were defined based on Albert and Miller (1996), as follows: The Glide phase begins when the vertical velocity of the centre of mass (COM) reaches −0.1 m/s and continues until the COM reaches its lowest point. The Transition phase starts at the end of the Glide and lasts until the vertical velocity of the right foot heel marker becomes

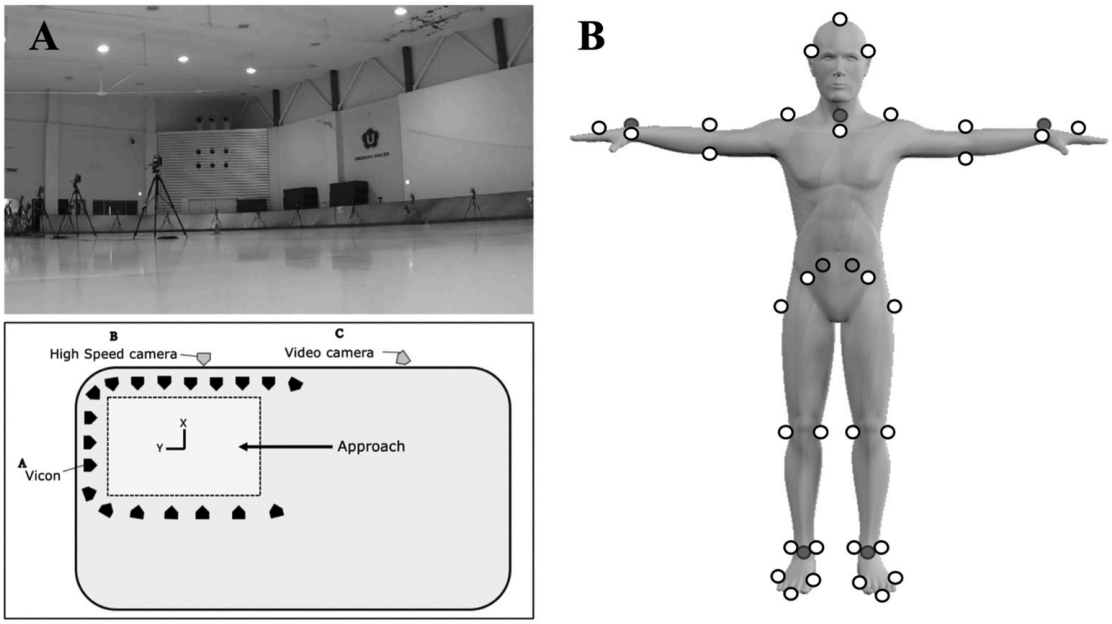


Figure 1. Experiment environment and reflective makers placement.

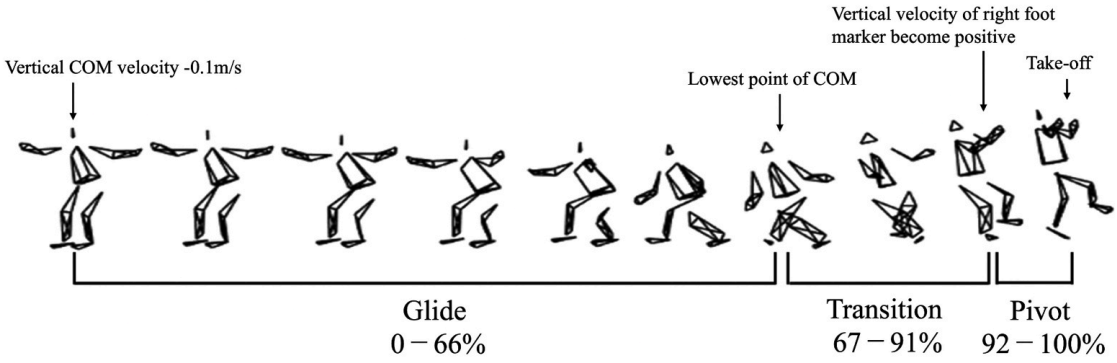


Figure 2. Analysis phases.

positive. This indicates that the skates begin to rock forward continuously towards the toe pick until they come into contact with the ice. The Pivot phase begins at the end of the Transition and continues until take-off. The duration of the phases was normalised based on their average times, with the Glide spanning 0–66%, the Transition covering 67–91%, and the Pivot extending from 92 to 100% of the take-off sequence (Figure 2).

Jump parameters

For the kinematic analysis, the skater's body was modelled as a system of 14 rigid, interconnected, and axisymmetric links. The centre of the wrist, elbow, ankle, and knee joints was defined as the midpoint between the two markers attached to the joint. The hip joint centre was determined using the method described by Kurabayashi et al. (2003). Body segment mass and moment of inertia were calculated using segmental coefficients obtained from young Japanese athletes (Ae et al., 1992). The mass of each skate was

assumed to be 1 kg and added to the foot mass, with the COM of the skate assumed to coincide with that of the foot. This value was based on the measured mass of one representative participant's skate.

To analyse the jump performance, the following variables were calculated: approach speed, duration of take-off sequences, average shoulder and pelvis rotational velocities (number of rotations of the line segments connecting the right and left markers during flight, divided by the duration of the flight phase), estimated jump height (derived from flight time using the equations of motion), measured jump height (difference between the COM height at take-off and its highest point during the jump), jump distance (COM's horizontal linear distance between its take-off and landing sites), vertical and horizontal velocities of the COM at take-off, and take-off angle (calculated as the inverse tangent of the vertical velocity divided by the horizontal velocity). Since previous studies used different methods for calculating jump height, both estimated and measured values were computed in the present study to ensure consistency in comparisons.

Angular momentum

In the absolute coordinate system, the whole-body angular momentum about the COM was computed following the procedure described by Tang (1995). The following formula was used to calculate the angular momentum components L_X, L_Y, L_z:

$$L_X = \sum m_i\{(Y_{ic} - Y_c)(V_{Zic} - V_{Zc}) - (Z_{ic} - Z_C)(V_{Yic} - V_{Yc})\} + \left\{\sum (A_i\omega_{Xi} + F_i\omega_{Yi} + E_i\omega_{Zi})\right\}$$

$$L_Y = \sum m_i\{(Z_{ic} - Z_c)(V_{Xic} - V_{Xc}) - (X_{ic} - X_C)(V_{Zic} - V_{Zc})\} + \left\{\sum (F_i\omega_{Xi} + B_i\omega_{Yi} + D_i\omega_{Zi})\right\}$$

$$L_Z = \sum m_i\{(X_{ic} - X_c)(V_{Yic} - V_{Yc}) - (Y_{ic} - Y_C)(V_{Xic} - V_{Xc})\} + \left\{\sum (E_i\omega_{Xi} + D_i\omega_{Yi} + C_i\omega_{Zi})\right\}$$

The coordinates of the COM of segment i are represented by X_{iC}, Y_{iC}, Z_{iC} the velocity components of the COM of segment i are indicated by $V_{Xic}, V_{YiC}, V_{Zic}$; and the coordinates of the whole-body COM are represented by X_C, Y_C, Z_C. D_i, E_i, F_i are the products of inertia (off-diagonal elements of the inertia tensor) written in the absolute coordinate system, while A_i, B_i, C_i are the primary moments of inertia of segment i. The angular velocity components of segment i with respect to its COM are $\omega_{Xi}, \omega_{Yi}, \omega_{Zi}$. By differentiating the unit vectors of the local coordinate system specified for each segment, the angular velocities of each segment were determined (Wadati, 1983).

Because the axis of rotation during jumps does not necessarily align with the absolute coordinate system due to body tilting and twisting, angular momentum components were transformed using direction cosines to align with the body's rotational axes. The rotational axes of the body were defined as follows: (1) The z-axis was

defined as the vector connecting the midpoint of both ASIS to the midpoint of both acromial processes. (2) The auxiliary vector (y') was defined as the vector from the midpoint of the ASIS to the left ASIS. (3) The x-axis was determined as the cross product of Y' and Z. (4) The y-axis was defined as the cross product of z and x. To examine the contributions of different body segments, angular momentum was analysed using both a three-segment model (arms, legs, and head-trunk) and a five-segment model (right arm, left arm, right leg, left leg, and head-trunk). Angular momentum values were normalised by dividing by the skater's mass with the squared height (Dapena, 1980; Hinrichs, 1987). For simplicity, the values were multiplied by 10^3, and the units are expressed as [$\times 10^{-3} s^{-1}$].

Statistical analysis

All variables are expressed as mean and standard deviation. The Shapiro—Wilk test was used to assess the normality of each dataset. Pearson's correlation coefficient was used to evaluate the relationship between whole-body angular momentum at take-off and jump parameters, and *p*-values were obtained to determine statistical significance. Additionally, one-way analysis of variance (ANOVA) was performed to test for significant differences in the maximum values of the angular momentum of each segment across the combined Glide and Transition phases. When a significant F-value was obtained, multiple comparisons were performed using the Bonferroni post hoc test. A *p*-value of <0.05 was considered statistically significant. All statistical analyses were performed using SPSS Statistics (version 29.0; SPSS Inc., Chicago, IL, USA).

Results

Table 1 presents the jump parameters and their correlation with whole-body angular momentum at take-off. No significant correlation was found between whole-body angular momentum at take-off (L_{BodyTO}: $154.8 \pm 17.0 \times 10^{-3} s^{-1}$) and any of the jump parameters.

Table 1. Comparison of jump parameters in T-LP and correlation analysis with whole-body angular momentum at take-off ($N = 10$).

	T-LP		
Metric		Correlation with L_body_{TO}	*p*-value
Angular momentum (L_body_{TO}) [$\times 10^{-3} s^{-1}$]	154.79 (±16.98)	–	–
Approach speed [m/s]	5.532 (±0.372)	.48	.16
Duration [s]	0.636 (±0.076)	.18	.62
Shoulder rotational velocity [rev/s]	4.067 (±0.161)	−.03	.94
Pelvis rotational velocity [rev/s]	4.249 (±0.133)	.18	.61
Flight time [s]	0.503 (±0.032)	.15	.68
Estimated jump height [m]	0.311 (±0.039)	.14	.70
Measured jump height [m]	0.269 (±0.038)	.11	.74
Jump distance [m]	1.441 (±0.182)	.07	.84
Vertical velocity at take-off [m/s]	2.368 (±0.197)	.01	.99
Horizontal velocity at take-off [m/s]	2.927 (±0.269)	−.01	.98
Take-off angle [deg]	39.02 (±2.602)	.01	.99

All values are expressed as mean (±SD).

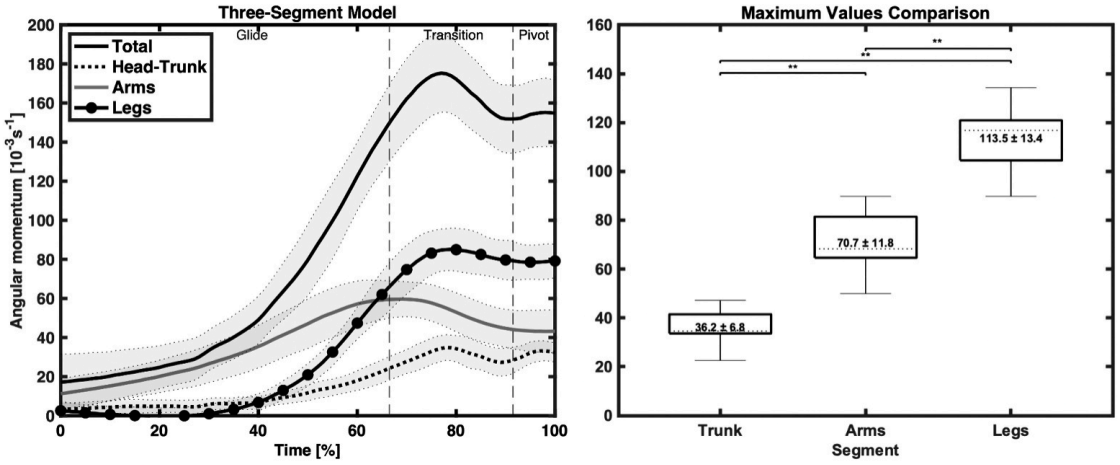


Figure 3. Angular momentum during the take-off phase, analysed using the three-segment model, and box plot of maximum values of each segment across the combined Glide and Transition phases. (**p <0.01, *p < 0.05).

Figure 3 shows the time-series changes in angular momentum during the take-off sequences, analysed using the three-segment model, as well as a box plot of the maximum angular momentum for each segment, with significant pairwise comparisons indicated by an asterisk (*). Whole-body angular momentum peaked at $176.5 \pm 20.7 \times 10^{-3}s^{-1}$ during the Transition (76.9% ± 2.0%). Both the arms and the legs also reached their peak during the Transition, with the arms peaking at $63.0 \pm 10.6 \times 10^{-3}s^{-1}$ (67.1% ± 9.1%) and the legs at $87.6 \pm 11.1 \times 10^{-3}s^{-1}$ (82.8% ± 7.3%). These results demonstrate that the maximum whole-body angular momentum is achieved during the Transition. Therefore, to clarify the contributions across the combined Glide and Transition phases, one-way ANOVA was conducted to analyse the maximum angular momentum obtained from each segment to examine differences between segments. The analysis revealed a significant main effect of segment ($F_{(2,18)} = 235.8$, $p < 0.001$). Bonferroni-adjusted post hoc comparisons indicated that significant differences were observed between the head-trunk and arms ($p < 0.01$), head-trunk and legs ($p < 0.01$), and arms and legs ($p < 0.01$).

Figure 4 shows the time-series changes in angular momentum during the take-off sequences, analysed using the five-segment model, as well as a box plot of the maximum angular momentum for each segment. The head-trunk segment exhibited a bimodal peak during the Transition (78.9% ± 2.8%) and Pivot (97.4% ± 2.6%), with maximum values of $36.2 \pm 6.5 \times 10^{-3}s^{-1}$ and $34.7 \pm 5.6 \times 10^{-3}s^{-1}$, respectively. The right arm peaked at $40.2 \pm 6.3 \times 10^{-3}s^{-1}$ during the Transition (72.8% ± 6.1%). In contrast, the left arm exhibited a bimodal increase, peaking at $28.1 \pm 7.1 \times 10^{-3}s^{-1}$ during the Glide and $28.0 \pm 8.9 \times 10^{-3}s^{-1}$ during the Pivot (99.2% ± 2.1%). The right leg peaked at $38.2 \pm 6.7 \times 10^{-3}s^{-1}$ during the Transition (68.3% ± 3.4%), while the left leg reached its peak at $75.4 \pm 9.0 \times 10^{-3}s^{-1}$ during the Transition (89.9% ± 2.4%). These findings indicate that the right arm, right leg, and left leg exhibited unimodal peaks during the Transition. In contrast, the head-trunk segment and the left arm exhibited bimodal peaks, indicating that maximum values do not necessarily occur during the Transition. To further examine the differences

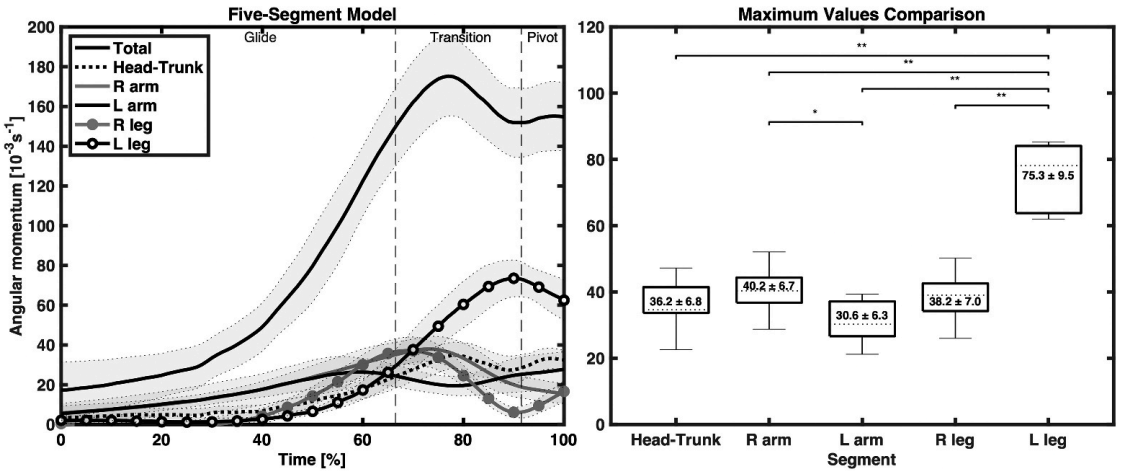


Figure 4. Angular momentum during the take-off phase, analysed using the five-segment model, and box plot of maximum values of each segment across the combined Glide and Transition phases.(**p <0.01, *p < 0.05).

in angular momentum among the segments a one-way ANOVA was conducted on the maximum angular momentum values obtained across the combined Glide and Transition phases. The analysis revealed a significant main effect of segment ($F_{(4,36)} = 98.8$, $p < 0.001$). Bonferroni-adjusted post hoc comparisons showed that significant differences were found between the right arm and left arm ($p = 0.03$), head-trunk and left leg ($p < 0.01$), right arm and left leg ($p < 0.01$), left arm and left leg ($p < 0.01$), and right leg and left leg ($p < 0.01$). In contrast, no significant differences were observed among the other segment pairs.

Discussion

This study aimed to investigate the specific characteristics of T-LP by comparing its fundamental kinematic variables with those of other jumps reported in previous studies. Additionally, this study sought to quantify the contributions of individual body segments to angular momentum generation during the take-off sequences, providing valuable insights into take-off techniques.

Jump parameters

Previous studies on the T-LP primarily focused on flight time and jump height. In this study, in addition to these parameters, fundamental kinematic quantities were calculated to identify the distinct characteristics of the T-LP compared with other jumps, particularly the axel, which has been the most extensively studied.

According to previous studies, the flight time for the T-LP has been reported as 0.488 ± 0.203 s for senior and junior Japanese national level skaters and 0.548 ± 0.268 s for Olympic skaters (Ikegami et al., 2004). The findings of this study (0.503 ± 0.032 s) showed a slightly longer flight time compared to national level skaters but did not reach the values reported for Olympic skaters. Similarly, the jump height was lower than that of

Olympic skaters (present study : 0.311 ± 0.039 m; Olympic skaters: 0.421 m; Sakurai et al., 1999). These results suggest that the skaters who participated in this study possess a high level of performance comparable to national level but fall slightly short of the performance level of international elite skaters.

When comparing T-LP with the triple axel (T-A), the T-LP exhibited a tendency towards shorter flight time (T-LP: 0.503 ± 0.032 s; T-A: 0.548–0.604 s; Ikegami et al., 2004) and lower jump height (T-LP: 0.269 ± 0.038 m; T-A: 0.351–0.383 m; Ikegami et al., 2004). Furthermore, the jump height of the T-LP was even lower than that reported for the double axel (D-A) performed by novice-level skaters (0.336 ± 0.043 m; Ridge et al., 2022). Additionally, the jump distance of the T-LP was shorter than that of the D-A performed at World Championships (T-LP: 1.441 ± 0.182 m; D-A: 2.245 ± 0.407 m; Hirosawa et al., 2022). On the other hand, the approach speed of the T-LP (5.532 ± 0.372 m/s) was faster than that reported for the T-A (4.90–5.12 m/s; Ikegami, 2004) and the D-A (4.5 ± 0.6 m/s; Albert & Miller, 1996), indicating one of the highest approach speeds among reported jumps. These findings suggest that, despite the high approach speed, T-LP is a compact jump characterised by relatively low height and short distance. Furthermore, the faster approach speed observed in this study was likely influenced by the experimental conditions. In previous studies, data were collected from competitions, where approach speeds may have been constrained by programme requirements. In contrast, this study was conducted under conditions in which skaters could freely accelerate without such constraints. This suggests that achieving high rotational jumps may not necessarily require a high approach speed.

In previous studies, the reference points used to evaluate rotational velocity during flight have varied. Some studies used the shoulder (Ikegami et al., 2004, 2005), while others focused on the pelvis (King et al., 2004; Ridge et al., 2022). Therefore, in the present study, rotational velocities were calculated at both the shoulder and pelvis and compared with previously reported values. The results indicated that the rotational velocity during flight was 4.067 ± 0.161 rev/s at the shoulder and 4.249 ± 0.133 rev/s at the pelvis. Both values were lower than those reported for T-A (4.34–4.53 rev/s; Ikegami et al., 2004) and the quad Salchow (5.04 rev/s; Ikegami et al., 2005). In contrast, it was higher than that of D-A (3.88 ± 0.17 rev/s; Ridge et al., 2022) and the triple toe-loop (3.8–3.9 rev/s; King et al., 2004). This trend suggests that jumps with higher rotational difficulty, such as the D-A, triple toe-loop, T-LP, T-A, and quad salchow, generally exhibit progressively higher rotational speed.

Whole-body angular momentum

The T-LP is categorised as an ‘edge jump’ because it takes off from a single edge of the skate blade, similar to the axel and salchow, which generate flight time and angular momentum using a single foot. In contrast, toe jumps, such as the toe-loop, flip, and lutz, incorporate a toe-pick thrust to aid the take-off. According to Sakurai et al. (1999), toe jumps tend to achieve greater jump heights than edge jumps, reflecting differences in their movement structures. In this study, whole-body angular momentum for the T-LP significantly increased during the latter half of the Glide and the first half of the Transition (Figure 3). This pattern is similar to the time-series changes observed in the double axel (Albert & Miller, 1996), suggesting a common angular momentum generation mechanism across

edge jumps. The maximum whole-body angular momentum ($176.5 \pm 20.7 \times 10^{-3}s^{-1}$) was achieved during the Transition. Following this peak, angular momentum decreased by approximately 12% ($21.7 \pm 26.8 \times 10^{-3}s^{-1}$) leading up to take-off. This reduction likely occurred due to changes in body posture and energy dispersion, hindering rotational movement. These findings align with previous observations in D-A (Albert & Miller, 1996).

Although research on angular momentum in figure skating is relatively limited, existing studies provide valuable comparative insights. Albert and Miller (1996) reported a whole-body angular momentum of $128.6 \pm 23.0 \times 10^{-3}s^{-1}$ for Olympic-level skaters performing D-A, whereas Ridge et al. (2022) observed that junior skaters executing the identical jump had a higher value of $153.0 \pm 61.7 \times 10^{-3}s^{-1}$ (Note: This is an estimated value standardised based on weight and height, considering the angular momentum described in the text.) This discrepancy, where Olympic-level skaters exhibited approximately $24.4 \times 10^{-3}s^{-1}$ lower angular momentum than their junior counterparts, suggests that differences in technical proficiency influence the mechanical characteristics of jump execution. Specifically, elite skaters may employ more efficient movement strategies to optimise rotational control while minimising excessive angular momentum, thereby achieving greater stability and consistency in jump performance.

Additionally, King et al. (2004) compared the triple toe-loop of male skaters who had mastered the quadruple toe-loop with those who had not. They found that skaters capable of executing quadruple toe-loops tended to achieve longer flight times. This suggests that as athletes refine their technique, the relative importance of angular momentum may diminish, with other factors, such as jump height and moment of inertia, compensating for this change. These findings emphasise that jump performance depends not only on angular momentum but also on a combination of technical and biomechanical factors. Elite-level skaters are likely to achieve more effective rotation by optimising their take-off mechanics and aerial positioning. This interpretation aligns with the results of our study, which showed no significant correlation between whole-body angular momentum at take-off and rotational speed or flight time. Based on these findings, it can be concluded that while angular momentum plays a crucial role in generating rotational speed, it is insufficient to determine jump performance or efficiency. Understanding these sports-specific characteristics is essential for developing a more comprehensive framework for analysing jump dynamics in figure skating.

Segmental angular momentum

In this study, angular momentum was analysed using both the three-segment and five-segment models. Time-series changes in angular momentum for each segment during take-off sequences were visualised to examine their contributions to angular momentum generation.

Previous research on longitudinal rotational movements in off-ice conditions has examined how individual body segments contribute to angular momentum generation. For example, Takeuchi et al. (2024) analysed national-level female skaters performing off-ice rotational jumps. They reported that the athletes achieved a rotation of 615.1 ± 51.6 degrees during a flight time of 0.47 ± 0.02 s, and the angular momentum about the vertical axis at take-off was $104 \pm 10 \times 10^{-3}s^{-1}$. Furthermore, it was noted that this angular momentum was primarily generated by the movements of the arms and trunk before take-off. In contrast,

the results of this study (Figure 3B) showed that, in the T-LP, the angular momentum of the upper limbs increased earlier during the Glide; however, during the Transition, the angular momentum of the lower limbs exceeded that of the upper limbs, with the non-supporting leg showing particularly notable contributions. Moreover, the whole-body angular momentum at take-off in the T-LP was greater than that reported for the off-ice rotational jumps. These findings suggest a fundamental difference in the angular momentum generation mechanisms between the T-LP and off-ice rotational jumps, indicating that the lower limbs may play a more dominant role in the T-LP. In other words, these results support the hypothesis of the present study that the lower limbs play a major role in angular momentum generation. This difference may be attributed to the unique movement structure of figure skating, in which the skater transitions from horizontal gliding to vertical take-off, necessitating greater control of angular momentum through the lower limbs.

In ballet pirouettes have shown that the contribution of the legs to angular momentum is relatively small, whereas the trailing arm (i.e., the arm opposite the direction of rotation) plays a primary role in generating angular momentum (Kim et al., 2014). Similarly, in the double-twisting backward somersault in gymnastics, the movement of the arms and upper body is crucial for generating angular momentum (Harigai et al., 2021). In particular, it has been reported that rapidly drawing the arm opposite to the twisting direction towards the chest is a key factor in angular momentum generation. Further analysis using the five-segment model revealed that the angular momentum generated by the right arm was greater than that of the left arm (Figure 4B). Because all skaters placed their left arm in front and their right arm to the side, the left arm functioned as the lead arm, whereas the right arm acted as the trailing arm. These results are consistent with previous studies on off-ice longitudinal rotational movements, in which the trailing arm played a major role in generating angular momentum.

Time-series analysis of angular momentum revealed distinct characteristics across different phases of the jump. During the Glide, all segments exhibited a coordinated increase in angular momentum, suggesting that the entire body was preparing to initiate rotation. In the Transition, all segments except the leading left arm reached their maximum angular momentum. Notably, the angular momentum of the supporting right leg began to decrease in the latter half of the Transition, while that of the non-supporting left leg increased. Furthermore, during the Pivot, the angular momentum of the non-supporting left leg decreased significantly, with a reduction of the trunk and other limbs. The Pivot, during which only the toe-pick remains in contact with the ice, represents the transition from the take-off posture to a compact air position. This phenomenon, whereby angular momentum decreases in one segment while increasing in another, is referred to as the 'transfer of angular momentum' (Ae & Fujii, 2002) and is regarded as a fundamental mechanism for efficient rotational motion. The transfer of angular momentum to the non-supporting left leg observed during the Transition may indicate reduce the mechanical load on the supporting leg, while simultaneously contributing to the maintenance of rotation and the achievement of sufficient flight time. Moreover, the redistribution of angular momentum to the trunk and other limbs observed during the Pivot is consistent with the findings of Albert and Miller (1996) for D-A. The Pivot, during which only the toe-pick remains in contact with the ice, represents the transition from the take-off posture

to a compact air position. This redistribution of angular momentum suggests the possibility a crucial role in stabilising rotational control in the air. These findings suggest a movement strategy aimed at efficiently transferring the energy generated during take-off sequence into air position, reflecting the high technical demands and the complexity of movement coordination required to execute T-LP.

Conclusion

This study examined the kinematic characteristics of the T-LP in figure skating, focusing on the angular momentum generation and its relationship with jump performance. The results showed that although the AM increased first due to the contribution of the upper limbs during the Glide, the AM of the lower limbs increased more significantly during the Transition, with a particularly notable contribution from the non-supporting left leg. This mechanism appears to be critical for the successful execution of T-LP and differs from off-ice rotational movements, where upper-limb motion plays a more dominant role. Additionally, no significant correlation was found between whole-body angular momentum at take-off and key jump parameters such as flight time, jump height, and rotational speed. This suggests that jump performance is not determined solely by angular momentum but rather by a combination of biomechanical factors, including lower limb mechanics, take-off technique, and aerial posture control.

Understanding the distinct angular momentum generation mechanisms in edge jumps, such as the T-LP, could aid in optimising training strategies and refining take-off techniques for figure skaters. This study visualised the time-series changes in angular momentum of individual body segments during the take-off sequence and quantitatively examined their contributions, thereby revealing the importance of the lower limbs and the strategic generation of angular momentum through the non-supporting leg in the T-LP. However, several limitations should be acknowledged. First, the participants were limited to 10 female skaters, and it is unclear whether the results can be generalised to a wider range of skill levels or male skaters. Additionally, individual variations in skating technique may have contributed to fluctuations in the calculated parameters. These limitations should be considered when interpreting the results, and future studies should include a more diverse sample to enhance the generalisability of the findings. Furthermore, although this study focused on analysing angular momentum generation mechanisms, it did not elucidate the specific movements that lead to these mechanisms. Identifying movement patterns that contribute to changes in angular momentum would provide practical insights for training guidance. Furthermore, further research is needed to compere successful jumps with unsuccessful jumps and to analyses skaters who are in the process of learning the T-LP. Such investigations are expected to elucidate key factors that differentiate between these attempts and elucidating the movement characteristics specific to each stage of skill acquisition. Nevertheless, this study is pioneering in that it focuses on loop jumps, which have been scarcely studied to date, and reveals the mechanisms of angular momentum generation. The findings are expected to contribute to future research that promotes the advancement of jump techniques in figure skating.

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